

Sessional Exam I

Date: 19/02/2025

Time: 2:00- 3:00 PM

Introduction to Data Science and Design Thinking (BCE-O633)

Note: Section A contains 4 questions of 6 marks each attempt any 2.

Section B contains 2 questions of 8 marks each, attempt any 1.

Section A

~~Q1.~~ Define the term Data Science, and explain steps to learn data science.

CO1

~~Q2.~~ Discuss the 5 characteristics of data

CO1

Q3. What are the stages in a data science project? explain each phase in brief.

CO2

Q4. What are data security issues in a data science project?

CO2

Section B

Q1. You own a bike share company and you want to increase your revenue in next semester. What will be your policy to achieve the goal? Discuss your approach with respect to different Data Science Roles and explain the task of each persona.

CO1

~~Q2.~~ What are the applications of Data Science in various fields? Mention any five applications with detail.

CO1

Introduction to Data Science and Design Thinking (BCE-O633)

Sessional Exam II

Date: 03/04/2025

Time: 2:00- 3:00 PM

Note: Section A contains 4 questions of 6 marks each attempt any 2.
Section B contains 2 questions of 8 marks each, attempt any 1.

Section A

S.No	Question	CO	BL
Q1.	What is data Collection? Discuss the different methods for primary Data Collection.	CO3	3
Q2.	What is Data Reduction? Discuss any two Data Reduction Strategies.	CO4	2
Q3.	A dataset has values 4, 8, 12, 16, 20, 24, 28. Construct a box plot by identifying the minimum, Q1, median, Q3, and maximum.	CO3	4
Q4.	Discuss data transformation strategies, define any five strategies in detail.	CO4	5

Section B

S.No	Question	CO	BL
Q1.	Explain the different steps involved in data pre-processing and discuss how each step contributes to improving data quality.	CO2	2
Q2.	A teacher recorded the test scores of 10 students: 55, 60, 65, 70, 70, 75, 80, 85, 90, 95. Calculate the mean, median, mode, and standard deviation.	CO3	3